Solutions :-

S, A, B, C, G are the nodes

Find the least cost path from S to G

Step 1 : Start at city S (initial cost)

Queue : [S, 0]

Step 2 : Expand city 'S'

Queue : [A, 4], [B, 2]

Step 3 : Expand city 'B'

Queue : [A, 3], [C, 8], [A, 4]

Step 4 : Expand city 'A'

Queue : [G, 12], [C, 8]

Step 5 : Expand city 'C'

Queue : [G, 13], [G, 12]

Step 6 : Expand city 'G'

City G is the goal node, where we have found the least cost path from S to G

The final path cost :

The least cost path from S to G is $S \rightarrow B \rightarrow A \rightarrow G$ with a total cost of 12.

Stack [D, G]

Visit G, no neighbours

Stack [D]

Visit D, push neighbours (I, J) onto the stack

Stack [J, I]

Visit I, no neighbours

Stack [J]

Visit J, push neighbours (L) onto the stack

Stack [L]

Visit L, no neighbours.

Stack []:

Resulting DFS Traversal : A - B - E - K - F - C - H - G - D - I - J - L

Visit D, push neighbours [I, J] onto the stack

Stack [J, I]

Visit I, no neighbours

Stack [J]

Visit J, push neighbours (L) onto the stack

Stack [L]

Visit L, no neighbours.

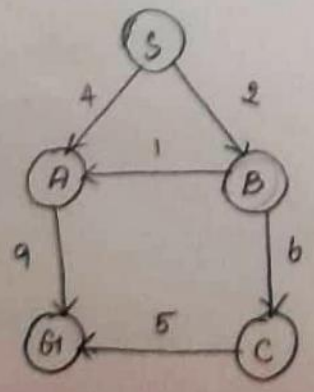
Stack []:

Resulting DFS Traversal : A - B - E - K - F - C - H - G - D - I - J - L

2) Uniform Cost Search (UCS)

From	To	Cost (meters)
S	A	4
S	B	2
A	G	9
B	A	1
B	C	6
C	G	5

Graph :-



Visit J, add its neighbours (L) to the queue

Queue [K, G, L]

Visit K, no neighbour nodes.

Queue [G, L]

Visit G, no neighbour nodes.

Queue [L]

Visit L, no neighbours node.

Queue[ ]:

Resulting BFS traversal : [A, B, C, D, E, F, H, I, J, K, G, L]

Depth First Search (DFS)

Start DFS from node A.

Stack [A]

Visit A, push neighbours (B, C, D) onto the stack

Stack [D, C, B]

Visit B, push neighbours (E, F) onto the stack

Stack [D, C, F, E]

Visit E, push neighbours (K) onto the stack

Stack [D, C, F, K]

Visit K, no neighbours

Stack [D, C, F]

Visit F, no neighbours.

Stack [D, C]

Visit C, push neighbour (H) onto the stack

Stack [D, H]

Visit H, push neighbour (G) onto the stack

Queue [J, K, G]

Visit J, add its neighbours (L) to the queue.

Queue [K, G, L]

Visit K, no neighbour nodes.

Queue [G, L]

Visit G, no neighbour nodes.

Queue [L]

Visit L, no neighbours node.

Queue[]:

Resulting BFS traversal : [A, B, C, D, E, F, H, I, J, K, G, L]

Depth First Search (DFS)

Start DFS from node A.

Stack [A]

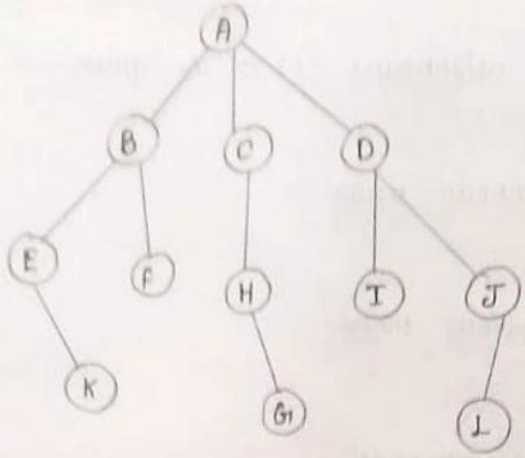
Visit A, push neighbours (B, C, D) onto the stack

Stack [D, C, B]

Visit B, push neighbours (E, F) onto the stack

Stack [D, C, F, E]

Visit E, push



Solution:-

Breadth First Search (BFS)

Steps of BFS (Starting from node A)

Queue [A]

Visit A, add its neighbours (B, C, D) to the queue.

Queue [B, C, D]

Visit B, add its neighbours (E, F) to the queue

Queue [C, D, E, F]

Visit C, add its neighbours (H) to the queue

Queue [D, E, F, H]

Visit D, add its neighbours (I, J) to the queue.

Queue [E, F, H, I, J]

Visit E, add its neighbours (K) to the queue.

Queue [F, H, I, J, K]

Visit F, no neighbour node.

Queue [H, I, J, K]

Visit H, add its neighbours (G) to the queue

Queue [I, J, K, G]

Visit I, no neighbours node.